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Past Papers Nat 5 Chemistry

2022 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/100)	%	
A	70+	70%	42.5%
B	58+	58%	20.6%
C	46+	46%	16.7%
D	34+	34%	11.9%
No award	<34	<34%	8.3%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	17.2 /25	45.8 /75	No Assignment in 2022

2022 National 5 Chemistry Marking Scheme

MC Qu	Answer	% Pupils Correct	Reasoning
1	C	91	☒ A Hydrogen H has an atomic number of 1. Hydrogen is a non-metal. ☒ B Arsenic As has an atomic number of 33. Arsenic is a non-metal. ☒ C Rhodium Rh has an atomic number of 45. Rhodium is a metal. ☒ D Radon Rn has an atomic number of 86. Radon is a non-metal.
2	B	83	☒ A Protons are positive and neutrons are neutral $\therefore$ overall this would be positive. ☒ B Atoms are neutral because no. of protons <u>equals</u> no of electrons ☒ C number of protons plus neutrons is greater than number of electrons ☒ D number of electrons plus protons is greater than number of neutrons
3	A	46	☒ A weak forces of attraction are found between molecules not inside molecules ☒ B strong forces of attraction are found inside molecules not between molecules ☒ C weak forces of attraction are found between molecules not inside molecules ☒ D strong forces of attraction are found inside molecules not between molecules
4	C	52	☒ A Adding more solvent would dilute the solution and decrease the concentration ☒ B Adding more solute would increase the concentration of the solute dissolved ☒ C Adding solute increases concentration. Adding solvent decreases concentration ☒ D Adding more solvent would dilute the solution and decrease the concentration
5	B	91	☒ A The shape is similar to the shape of CH_4 and is called tetrahedral ☒ B The shape is similar to the shape of H_2O and is called angular ☒ C The shape is similar to the shape of HCl and is called linear ☒ D The shape is similar to the shape of NH_3 and is called trigonal pyramidal
6	A	91	☒ A Electronegativity of O=3.4 & Electronegativity of H=2.2 $\therefore$ Electronegativity Difference = 1.2 ☒ B Electronegativity of N=3.0 & Electronegativity of H=2.2 $\therefore$ Electronegativity Difference = 0.8 ☒ C Electronegativity of C=2.6 & Electronegativity of H=2.2 $\therefore$ Electronegativity Difference = 0.4 ☒ D Electronegativity of C=2.6 & Electronegativity of O=3.4 $\therefore$ Electronegativity Difference = 0.8
7	A	54	☒ A Copper forms at negative electrode and chlorine gas forms at positive electrode ☒ B Copper forms at negative electrode as positive Cu^{2+} ions move to negative electrode ☒ C Chlorine gas forms at positive electrode as negative Cl^- ions move to positive electrode ☒ D Chlorine gas forms at positive electrode as negative Cl^- ions move to positive electrode
8	D	35	☒ A Calcium oxide cannot be formed by the neutralisation of an acid ☒ B Hydrogen nitrate cannot be formed by the neutralisation of an acid ☒ C Sodium hydroxide cannot be formed by the neutralisation of an acid. ☒ D Potassium ethanoate is a salt formed by the neutralisation of ethanoic acid by a base like sodium hydroxide
9	D	54	☒ A pH=3 is acidic and ammonia dissolves in water to form an alkali with pH>7 ☒ B pH=5 is acidic and ammonia dissolves in water to form an alkali with pH>7 ☒ C pH=7 is acidic and ammonia dissolves in water to form an alkali with pH>7 ☒ D pH=9 is alkaline and ammonia dissolves in water to form an alkali with pH>7
10	C	48	☒ A Methane CH_4 burns to form CO_2 and H_2O . The CO_2 would turn limewater milky. ☒ B Carbon Monoxide CO burns to form CO_2 . The CO_2 would turn limewater milky. ☒ C Hydrogen H_2 burn to form H_2O only. H_2O would condense as a colourless liquid. ☒ D Ethane C_2H_6 burns to form CO_2 and H_2O . The CO_2 would turn limewater milky.
11	C	73	☒ A C_4H_{10} molecule is butane and has a boiling point of -1°C ☒ B C_4H_8 molecule is but-1-ene and has a boiling point of -6°C ☒ C $\text{C}_3\text{H}_7\text{COOH}$ molecule is butanoic acid and has a boiling point of 164°C ☒ D $\text{C}_4\text{H}_9\text{OH}$ molecule is butan-2-ol and has a boiling point of 100°C
12	D	52	☒ A molecule has no $\text{C}=\text{C}$ double bond and would not decolourise bromine solution ☒ B molecule has no $\text{C}=\text{C}$ double bond and would not decolourise bromine solution ☒ C molecule has no COOH Carboxyl group and would not produce an acidic pH ☒ D Carboxyl $-\text{COOH}$ group has acid pH & $\text{C}=\text{C}$ double bond decolourises bromine solution

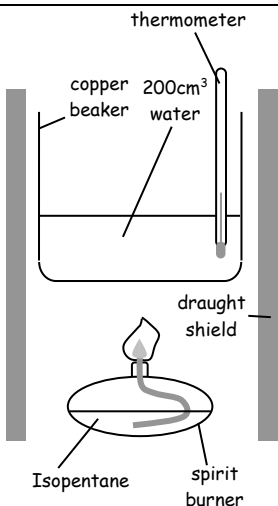
13	B	79	Structure					
			Formula	C ₃ H ₆ O	C ₄ H ₈ O	C ₅ H ₁₀ O		
			Relationship	If n=3 then 2n=6	If n=4 then 2n=8	If n=5 then 2n=10		
			General Formula	C _n H _{2n} O	C _n H _{2n} O	C _n H _{2n} O		
14	B	59	Cycloalkanes have a general formula of C _n H _{2n} .					
			no of Carbons C _n	C ₃	C ₄	C ₅		
			Formula C _n H _{2n}	C ₃ H ₆	C ₄ H ₈	C ₅ H ₁₀		
			gfm	(3x12)+(6x1) = 42	(4x12)+(8x1) = 56	(5x12)+(10x1) = 70		
15	A	84	☒ A metallic bonding: attraction between positive ions and delocalised electrons					
			☒ B the positive ion in metallic bonding is formed by the nucleus and the inner shells					
			☒ C ionic bonding: attraction between positive ions and negative ions					
			☒ D covalent bonding: attraction between a shared pair of electrons and two nuclei					
16	B	86	☒ A melting point of 98°C is below the required 600°C					
			☒ B Melting point above 600°C and density below 3g cm ⁻³					
			☒ C density of 6.52 g cm ⁻³ is above the required density of 3g cm ⁻³					
			☒ D density of 8.96 g cm ⁻³ is above the required density of 3g cm ⁻³					
17	A	58	☒ A aluminium is extracted from ore by electrolysis and Al ₂ O ₃ is insoluble in water					
			☒ B calcium oxide is soluble in water and forms an alkali in water					
			☒ C copper metal is extracted by heating copper ore with carbon/carbon monoxide					
			☒ D lead metal is extracted by heating lead ore with carbon/carbon monoxide					
18	D	73	☒ A electrons travel through connecting wires not the electrolyte solution					
			☒ B electrons travel through connecting wires not the electrolyte solution					
			☒ C electrons travel from aluminium to nickel as aluminium is higher than nickel in ECS					
			☒ D electrons travel from aluminium to nickel through the connecting wires as aluminium is higher than nickel in ECS					
19	B	77	Electrochemical Series	Magnesium	Zinc (D)	Iron (A)	Tin (C)	Lead (B)
			Voltage					
20	C	81	$\begin{array}{l} \textcircled{1} \quad Br_2(l) \quad \quad \quad + \quad 2e^- \quad \longrightarrow \quad 2Br^-(aq) \\ \textcircled{2} \quad \quad \quad SO_3^{2-}(aq) + H_2O(l) \quad \longrightarrow \quad SO_4^{2-}(aq) + 2H^+(aq) + 2e^- \\ \text{Add} \\ \textcircled{1+2} \quad Br_2(l) + SO_3^{2-}(aq) + H_2O(l) + 2e^- \longrightarrow 2Br^-(aq) + SO_4^{2-}(aq) + 2H^+(aq) + 2e^- \\ \text{cancel} \\ \text{down} \quad Br_2(l) + SO_3^{2-}(aq) + H_2O(l) + \cancel{2e^-} \longrightarrow 2Br^-(aq) + SO_4^{2-}(aq) + 2H^+(aq) + \cancel{2e^-} \\ \text{redox} \quad Br_2(l) + SO_3^{2-}(aq) + H_2O(l) \longrightarrow 2Br^-(aq) + SO_4^{2-}(aq) + 2H^+(aq) \end{array}$					
21	A	82	Monomer	Repeating Unit	Polymer			
22	B	62	☒ A iron is the catalyst in the Haber Process not the Ostwald Process					
			☒ B platinum is the catalyst in the Ostwald process which produces nitric acid HNO ₃					
			☒ C iron is the catalyst in the Haber Process not the Ostwald Process					
			☒ D ammonia NH ₃ is the product of the Haber Process not the Ostwald Process					

23	C	67	☒ A All ^{222}Rn atom have same half-life due to having the same proton : neutron ratio ☒ B All ^{222}Rn atom have same half-life due to having the same proton : neutron ratio ☒ C ^{222}Rn has p:n ratio of 136:86 and the half-life is the same for all atoms of ^{222}Rn ☒ D The intensity of the radiation would change by having different size plants but the time taken for the radiation to halve (half-life) would remain the same.
24	D	55	☒ A alpha particles are stopped by paper $\therefore$ would not be able to escape through skin ☒ B long half-life would result in radiation escaping for potentially years to come. ☒ C alpha particles are stopped by paper $\therefore$ would not be able to escape through skin ☒ D an isotope with beta particles released which are able to escape the skin and a short half-life is the best combination for this treatment.
25	C	80	☒ A Beaker is an inaccurate method to measure volume. ☒ B Measuring cylinder is not as accurate as a 25cm^3 pipette for measuring volume ☒ C Most accurate method for measuring 25cm^3 is to use a 25cm^3 pipette. ☒ D Conical flask is an inaccurate method of measuring volume as it has no markings

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Long Qu	Answer	Reasoning																
1a(i)	Beta	Radiation	Alpha	Beta	Gamma													
		Mass	4	0	No mass													
		Charge	2	-1	No charge													
		Stopped by	Paper	Aluminium	Thick lead													
		Deflection	Towards negative	Towards positive	No defection													
		Use	Smoke detectors	Measuring thickness of paper in paper mill	Radiotherapy cancer treatment													
1a(ii)	Xenon	$^{131}_{53}\text{I} \rightarrow ^0_{-1}\text{e} + ^{131}_{54}\text{Xe}$																
1b(i)	15 days	<table><tr><th>Fraction</th><th>Number of half-lives</th></tr><tr><td>1</td><td>0</td></tr><tr><td>$\frac{1}{2}$</td><td>1</td></tr><tr><td>$\frac{1}{4}$</td><td>2</td></tr><tr><td>$\frac{1}{8}$</td><td>3</td></tr></table>		Fraction	Number of half-lives	1	0	$\frac{1}{2}$	1	$\frac{1}{4}$	2	$\frac{1}{8}$	3	1 half-life = 5 days 3 half-lives = 15 days				
		Fraction	Number of half-lives															
		1	0															
		$\frac{1}{2}$	1															
		$\frac{1}{4}$	2															
$\frac{1}{8}$	3																	
1b(ii)	Caesium-137	Caesium-137 levels would take 30 years to reach half of its original value and some caesium-137 would remain in the environments for over 200 years. The other radioisotopes have a half-life of in the days and the vast majority of the other radioisotopes would have decayed over 2-3 months.																
2a	Graph showing:	<table><tr><th>1 mark</th><th>1 mark</th></tr><tr><td>The axes of the graph have suitable labels and units.</td><td>All data points plotted accurately (within a half box tolerance) with either a line of best fit drawn or plots joined. This mark can only be accessed if linear scales for both axes have been provided.</td></tr></table>		1 mark	1 mark	The axes of the graph have suitable labels and units.	All data points plotted accurately (within a half box tolerance) with either a line of best fit drawn or plots joined. This mark can only be accessed if linear scales for both axes have been provided.											
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2b	5.33	$\text{Rate} = \frac{\Delta\text{Quantity}}{\Delta\text{Time}} = \frac{48 - 32}{4 - 1} = 5.33 \text{ cm}^3 \text{ min}^{-1}$																
2c	Mass of flask + contents	The flask and contents will decrease in mass as the hydrogen gas escapes from the flask. Mass can then be used to measure the rate of reaction by using change in mass as $\Delta\text{Quantity}$ in the above rate equation.																
2d	48	Using the same mass of calcium and the same volume of water means the same volume of hydrogen gas will be produced. The increased temperature will increase the reaction rate but it will still produce the same 48cm^3 but in less time.																
3a	ammonia and carbon dioxide	Problem Solving: gathering information from a passage																
3b	1.625	$32.5\% \text{ of } 5\text{kg} = \frac{32.5}{100} \times 5\text{kg} = 1.625\text{kg}$																
3c	<table><tr><td>hydrogen</td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td>hydroxide</td></tr></table>		hydrogen					hydroxide	<table><tr><th>Solution Type</th><th>Description</th></tr><tr><td>Acidic Solution</td><td>Concentration of H^+ ions > Concentration of OH^- ions</td></tr><tr><td>Neutral Solution</td><td>Concentration of H^+ ions = Concentration of OH^- ions</td></tr><tr><td>Alkaline Solution</td><td>Concentration of H^+ ions < Concentration of OH^- ions</td></tr></table>		Solution Type	Description	Acidic Solution	Concentration of H^+ ions > Concentration of OH^- ions	Neutral Solution	Concentration of H^+ ions = Concentration of OH^- ions	Alkaline Solution	Concentration of H^+ ions < Concentration of OH^- ions
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3d	Not toxic or flammable	Problem Solving: gathering information from a passage																
3e(i)	Contains two elements essential to healthy plant growth	Nitrogen, Phosphorus and Potassium are the three elements essential for healthy plant growth. Diammonium hydrogen phosphate $(\text{NH}_4)_2\text{HPO}_4$ contains two from nitrogen, phosphorus and potassium. A single nutrient fertiliser would contain only one of those three elements.																

3e(ii)	21.2	gfm (NH ₄) ₂ HPO ₄ : (2×14) + (8×1) + (1×1) + (1×31) + (4×16) = 132g (1 mark) $\% \text{ N} = \frac{(2 \times 14)}{132} \times 100 = 21.2\% \text{ (1 mark)}$																						
4a(i)	addition	HBr molecule adds across the C=C double bond by addition reaction. <table><tr><td colspan="5">Molecules which add across a C=C double bond to form one product</td><td colspan="5">Molecules which add across a C=C double bond to usually form two product</td></tr><tr><td>F₂</td><td>Cl₂</td><td>Br₂</td><td>I₂</td><td>H₂</td><td>H₂O</td><td>HF</td><td>HCl</td><td>HBr</td><td>HI</td></tr></table>			Molecules which add across a C=C double bond to form one product					Molecules which add across a C=C double bond to usually form two product					F ₂	Cl ₂	Br ₂	I ₂	H ₂	H ₂ O	HF	HCl	HBr	HI
Molecules which add across a C=C double bond to form one product					Molecules which add across a C=C double bond to usually form two product																			
F ₂	Cl ₂	Br ₂	I ₂	H ₂	H ₂ O	HF	HCl	HBr	HI															
4a(ii)																								
4a(iii)	HCl or hydrogen chloride																							
4b(i)																								
4b(ii)	Vinegar	Pure ethanoic acid (old name acetic acid) is diluted in water to make vinegar.																						
5	Open Ended Question: <table><tr><th colspan="4">Candidate Marks</th></tr><tr><td>0 marks</td><td>1 mark</td><td>2 marks</td><td>3 marks</td></tr><tr><td>22%</td><td>34%</td><td>29%</td><td>15%</td></tr></table>	Candidate Marks				0 marks	1 mark	2 marks	3 marks	22%	34%	29%	15%	3 mark answer	2 mark answer	1 mark answer								
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Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.																						
6a	2-methylbutane	2-methylbutane <table><tr><td>Methyl -CH₃ group on C₂</td><td>Longest carbon chain has four carbons (no C=C double bonds)</td></tr></table>			Methyl -CH ₃ group on C ₂	Longest carbon chain has four carbons (no C=C double bonds)																		
Methyl -CH ₃ group on C ₂	Longest carbon chain has four carbons (no C=C double bonds)																							
6b(i)	Fuel	A fuel is a substance which burns to release heat energy																						
6b(ii)	$\text{C}_5\text{H}_{12} + 8\text{O}_2$ ↓ $5\text{CO}_2 + 6\text{H}_2\text{O}$	$\text{C}_5\text{H}_{12} + 8\text{O}_2 \longrightarrow 5\text{CO}_2 + 6\text{H}_2\text{O}$																						
6b(iii)	10.032	heat energy = specific heat capacity × mass × change in Temperature $E_h = c \times m \times \Delta T$ $E_h = 4.18 \text{ kJ kg}^{-1} \text{ }^\circ\text{C}^{-1} \times 0.2\text{kg} \times 12^\circ\text{C}$ $E_h = 10.032 \text{ kJ}$																						

6b(iv)																																						
7a(i)	43.9	gfm NaCl = (1×23)+(1×35.5) = 23+35.5 = 58.5g no. of mol = volume × concentration = 0.5 litres × 1.5 mol l⁻¹ = 0.75mol mass = no. of mol × gfm = 0.75mol × 58.5g mol⁻¹ = 43.9g																																				
7a(ii)	Balance or Weighing Bottle																																					
7b(i)	106.7	Average Volume = $\frac{105 + 107 + 108}{3} = \frac{320}{3} = 106.7^{\circ}\text{C}$																																				
7b(ii)	Table showing:	<table><tr><th>Concentration (mol l⁻¹)</th><th>Boiling Point (°C)</th></tr><tr><td>0.5</td><td>101.3</td></tr><tr><td>1.0</td><td>104.0</td></tr><tr><td>1.5</td><td>106.7</td></tr></table>	Concentration (mol l ⁻¹)	Boiling Point (°C)	0.5	101.3	1.0	104.0	1.5	106.7																												
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7c	Line Graph																																					
7d	Increase in concentration, increase in boiling point	Problem Solving: Drawing conclusion from table of information																																				
8a	sulphuric acid	<table><tr><td>acid</td><td>+</td><td>metal oxide</td><td>→</td><td>salt</td><td>+</td><td>water</td></tr><tr><td>sulphuric acid</td><td>+</td><td>copper oxide</td><td>→</td><td>copper sulphate</td><td>+</td><td>water</td></tr><tr><td>acid</td><td>+</td><td>metal hydroxide</td><td>→</td><td>salt</td><td>+</td><td>water</td></tr><tr><td>sulphuric acid</td><td>+</td><td>barium hydroxide</td><td>→</td><td>barium sulphate</td><td>+</td><td>water</td></tr></table>	acid	+	metal oxide	→	salt	+	water	sulphuric acid	+	copper oxide	→	copper sulphate	+	water	acid	+	metal hydroxide	→	salt	+	water	sulphuric acid	+	barium hydroxide	→	barium sulphate	+	water								
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8b	Hydrogen	<table><tr><td>acid</td><td>+</td><td>metal carbonate</td><td>→</td><td>salt</td><td>+</td><td>water</td><td>+</td><td>carbon dioxide</td></tr><tr><td>sulphuric acid</td><td>+</td><td>sodium carbonate</td><td>→</td><td>sodium sulphate</td><td>+</td><td>water</td><td>+</td><td>carbon dioxide</td></tr><tr><td>acid</td><td>+</td><td>metal</td><td>→</td><td>salt</td><td>+</td><td>hydrogen</td><td></td><td></td></tr><tr><td>sulphuric acid</td><td>+</td><td>magnesium</td><td>→</td><td>magnesium sulphate</td><td>+</td><td>hydrogen</td><td></td><td></td></tr></table>	acid	+	metal carbonate	→	salt	+	water	+	carbon dioxide	sulphuric acid	+	sodium carbonate	→	sodium sulphate	+	water	+	carbon dioxide	acid	+	metal	→	salt	+	hydrogen			sulphuric acid	+	magnesium	→	magnesium sulphate	+	hydrogen		
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8c	Don't start to fizz when more carbonate is added	Sodium carbonate will react with sulphuric acid to form sodium sulphate, water and carbon dioxide. When the carbonate runs out, the fizzing will stop. More carbonate is added and stirred until the fizzing stops again. This is repeated until adding carbonate does not start to fizz showing that there is no acid remaining.																																				
8d	barium sulphate	<table><tr><td>Salt</td><td>copper sulphate</td><td>barium sulphate</td><td>sodium sulphate</td><td>magnesium sulphate</td></tr><tr><td>Solubility</td><td>soluble</td><td>insoluble</td><td>soluble</td><td>soluble</td></tr></table>	Salt	copper sulphate	barium sulphate	sodium sulphate	magnesium sulphate	Solubility	soluble	insoluble	soluble	soluble																										
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Solubility	soluble	insoluble	soluble	soluble																																		
8e	Reaction of acids to form water	A neutralisation reaction is the reaction of an acid with a base to form water. A salt is also produced in the reaction.																																				

9a	Answer including one from:	Same Atomic number Number of protons	Different Mass number Number of neutrons													
9b	Full outer shell	Atoms with a full outer shell are very stable and therefore unreactive.														
		Element	Helium	Neon	Argon	Krypton	Xenon	Radon								
		Electron Arrangement	2	2,8	2,8,8	2,8,18,8	2,8,18,18,8	2,8,18,32,18,8								
9c(i)	Equation showing:	$\text{XeF}_2 + \text{F}_2 \longrightarrow \text{XeF}_6$														
9c(ii)	covalent molecular	Xenon hexafluoride is covalent as it contains only non-metals atoms in its structure. The melting point of 49°C indicates that weak intermolecular attractions are found between molecules giving a covalent molecular structure. The melting point is too low for a covalent network structure.														
9c(iii)A	35	Catalysts speed up reactions without being used up in that reaction. The same mass of catalyst at the start remains at the end.														
9c(iii)B	£277.60	35g of catalyst must be obtained from four 10g tubs. 1 tub catalyst = £69.40 ∴ 4 tubs catalyst = £69.40 × 4 = £277.60														
10a	Answer containing:	Family of compounds with similar/same chemical properties a general formula														
10b(i)	Hydroxyl group	<table><tr><td>$-\text{O}-\text{H}$</td><td>$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array}$</td></tr><tr><td>hydroxyl group</td><td>carboxyl group</td></tr></table>							$-\text{O}-\text{H}$	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array}$	hydroxyl group	carboxyl group				
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hydroxyl group	carboxyl group															
10b(ii)	Secondary	<table><tr><td>$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{CH}_3 & & \text{H} \\ & & & & & & & \\ \text{H} & -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\ & & & & & & & \\ & \text{H} & & \text{OH} & & \text{H} & & \text{H} \end{array}$</td><td colspan="3">The circled carbon shown has attached to it: • 2 carbons • 1 hydrogen • 1 oxygen </td></tr></table>							$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{CH}_3 & & \text{H} \\ & & & & & & & \\ \text{H} & -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\ & & & & & & & \\ & \text{H} & & \text{OH} & & \text{H} & & \text{H} \end{array}$	The circled carbon shown has attached to it: • 2 carbons • 1 hydrogen • 1 oxygen						
$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{CH}_3 & & \text{H} \\ & & & & & & & \\ \text{H} & -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\ & & & & & & & \\ & \text{H} & & \text{OH} & & \text{H} & & \text{H} \end{array}$	The circled carbon shown has attached to it: • 2 carbons • 1 hydrogen • 1 oxygen															
10b(iii)	One structure from:	Structure		Name		Number of hydrogens attached to carbon with functional group		Type of Alcohol								
		$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{H} & & \text{H} \\ & & & & & & & & & \\ \text{H} & -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{OH} \\ & & & & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & & \text{H} & & \text{H} \end{array}$		pentan-1-ol		2		primary								
		$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{H} \\ & & & & & & & \\ \text{H} & -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{OH} \\ & & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & & \text{H} \\ & & & & & & & \\ & & & & & & & \text{H} \end{array}$		2-methylbutan-1-ol		2		primary								
		$\begin{array}{ccccccc} & & & \text{H} & & & & \\ & & & & & & & \\ & \text{H} & & \text{C} & & \text{H} & & \text{H} \\ & & & & & & & \\ \text{H} & -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{OH} \\ & & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & & \text{H} \end{array}$		3-methylbutan-1-ol		2		primary								
		$\begin{array}{ccccccc} & & & \text{H} & & & & \\ & & & & & & & \\ & \text{H} & & \text{C} & & \text{H} & & \text{H} \\ & & & & & & & \\ \text{H} & -\text{C} & - & \text{C} & - & \text{C} & - & \text{OH} \\ & & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & & \text{H} \\ & & & & & & & \\ & & & & & & & \text{H} \end{array}$		2,2-dimethylbutan-1-ol		2		primary								
		$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{OH} & & \text{H} \\ & & & & & & & \\ \text{H} & -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\ & & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & & \text{H} \\ & & & & & & & \\ & & & & & & & \text{H} \end{array}$		2-methylbutan-2-ol		0		tertiary								
11a(i)	Relights a glowing splint	<table><tr><td>Gas</td><td>Oxygen</td><td>Hydrogen</td><td>Carbon Dioxide</td></tr><tr><td>Gas Test</td><td>relights a glowing splint</td><td>burns with a pop</td><td>turns lime water milky</td></tr></table>							Gas	Oxygen	Hydrogen	Carbon Dioxide	Gas Test	relights a glowing splint	burns with a pop	turns lime water milky
Gas	Oxygen	Hydrogen	Carbon Dioxide													
Gas Test	relights a glowing splint	burns with a pop	turns lime water milky													

11a(ii)	$K^+ClO_3^-$	K is in group 1 and forms K^+ ions. Negative ion must balance the charge of the positive ion $\therefore ClO_3^-$									
11b(i)	Releases energy to surroundings	Type of Reaction		Definition							
		Exothermic		Reaction which releases energy to surroundings							
		Endothermic		Reaction which takes in energy from the surroundings							
11b(ii)	Potassium	Element	Ion	Flame Colour		Element	Ion	Flame Colour			
		Barium	Ba^{2+}	Green		Copper	Cu^{2+}	Blue-green			
		Potassium	K^+	Lilac		Strontium	Sr^{2+}	Red			
		Calcium	Ca^{2+}	Orange-red		Lithium	Li^+	Red			
		Sodium	Na^+	Yellow							
11c	2.4	gfm $C_6H_{12}O_6 = (6 \times 12) + (12 \times 1) + (6 \times 16) = 72 + 12 + 96 = 180g$									
		$\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{2.25}{180} = 0.0125\text{mol}$									
		$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ 1mol 6mol 0.0125mol 0.075mol									
		gfm $O_2 = 2 \times 16 = 32g$									
		mass = no. of mol $\times$ gfm = $0.075 \times 32 = 2.4g$									
12a(i)	decreases	Element		Na	Mg	Al	Si	P	S	Cl	
		Covalent Radius (pm)		160	140	124	114	109	104	100	
12a(ii)	increases	Group 1 Element		H	Li	Na	K	Rb			
		Covalent Radius (pm)		32	130	160	200	215			
12a(iii)	201	Group 2 Element		Be	Mg	Ca	Sr				
		Covalent Radius (pm)		99	140	174	-				
		Difference			41	34	27				
		Prediction		-	-	-	201				
12b	234	Covalent Radius of Br = 117 picometres (from table) Distance between nuclei = $2 \times \text{covalent radius} = 2 \times 117\text{pm} = 234\text{pm}$									
12c(i)	2,8	sodium atom Na = 2,8,1 $\therefore$ sodium ion $Na^+ = 2,8$									
12c(ii)	Na^+ has less occupied electron shells	Particle		Na atom	Na ⁺ ion						
		Electron Arrangement		2,8,1	2,8						
		No of occupied electron shells		3	2						
13	Open Ended Question:	3 mark answer				2 mark answer				1 mark answer	
		Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.				Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.				Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.	
		Candidate Marks									
		0 marks	1 mark	2 marks	3 marks						
		24%	38%	24%	14%						

2022 National 5 Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	C	5	3	91	1	1	0
2	B	9	83	4	3	0	0
3	A	46	10	28	16	1	0
4	C	35	5	52	8	1	0
5	B	1	91	3	5	0	0
6	A	91	1	2	5	1	0
7	A	54	4	33	8	1	0
8	D	10	28	27	35	1	0
9	D	9	20	16	54	1	0
10	C	14	27	48	10	1	0
11	C	3	8	73	15	1	0
12	D	15	9	23	52	1	0
13	B	5	79	10	6	1	0
14	B	13	59	16	11	1	0
15	A	84	4	6	5	1	0
16	B	11	86	2	1	1	0
17	A	58	17	18	6	1	0
18	D	14	4	8	73	0	0
19	B	4	77	3	16	1	0
20	C	9	7	81	3	1	0
21	A	82	4	3	11	0	0
22	B	7	62	10	21	1	0
23	C	16	8	67	8	1	0
24	D	8	18	18	55	1	0
25	C	3	12	80	4	0	0

2022 Nat5 Chemistry Section 2 Marks Data Analysis

2022 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.85	85%	46	2%
Q1aii	1	0.59	59%	47	3%
Q1bi	2	1.53	76%	61	3%
Q1bii	1	0.95	95%	27	1%
Q2a	2	1.23	62%	25	0%
Q2b	2	1.56	78%	43	1%
Q2c	1	0.17	17%	30	8%
Q2d	1	0.60	60%	41	3%
Q3a	1	0.92	92%	16	0%
Q3b	1	0.60	60%	52	7%
Q3c	1	0.60	60%	19	0%
Q3d	1	0.91	91%	22	0%
Q3ei	1	0.62	62%	50	3%
Q3eii	3	2.29	76%	57	3%
Q4ai	1	0.63	63%	28	3%
Q4aii	1	0.72	72%	48	3%
Q4aiii	1	0.26	26%	38	1%
Q4bi	1	0.64	64%	44	6%
Q4bii	1	0.57	57%	42	6%
Q5	3	1.38	46%	58	5%
Q6a	1	0.38	38%	51	4%
Q6bi	1	0.25	25%	33	3%
Q6bii	1	0.67	67%	54	2%
Q6biii	3	2.17	72%	66	5%
Q6biv	2	1.13	56%	47	9%
Q7ai	3	2.02	67%	67	5%
Q7aii	1	0.60	60%	39	5%
Q7bi	1	0.47	47%	40	2%
Q7bii	2	1.42	71%	58	4%
Q7c	1	0.67	67%	4	1%
Q7d	1	0.86	86%	46	3%
Q8a	1	0.65	65%	55	3%
Q8b	1	0.53	53%	50	3%
Q8c	1	0.35	35%	33	4%
Q8d	1	0.67	67%	58	6%
Q8e	1	0.32	32%	43	3%
Q9a	1	0.72	72%	41	2%
Q9b	1	0.71	71%	52	4%
Q9ci	1	0.13	13%	39	6%
Q9cii	1	0.43	43%	45	7%
Q9ciiiA	1	0.72	72%	46	6%
Q9ciiiB	1	0.43	43%	29	5%
Q10a	1	0.62	62%	51	4%
Q10bi	1	0.69	69%	57	3%
Q10bii	1	0.52	52%	48	9%
Q10biii	1	0.43	43%	48	6%
Q11ai	1	0.58	58%	43	6%
Q11aii	1	0.19	19%	36	9%
Q11bi	1	0.90	90%	33	2%
Q11bii	1	0.91	91%	23	1%
Q11c	3	1.61	54%	68	8%
Q12ai	1	0.88	88%	28	2%
Q12aii	1	0.80	80%	33	3%
Q12aiii	1	0.86	86%	32	4%
Q12b	1	0.64	64%	49	9%
Q12ci	1	0.64	64%	52	4%
Q12cii	1	0.33	33%	49	8%
Q13	3	1.29	43%	64	10%
Paper	75	45.81	61%		

2022 Nat5 Chemistry Section 2 Distribution of Marks Analysis

2022 Item	0	1	2	3
Q1ai	15	85		
Q1aii	41	59		
Q1bi	17	14	69	
Q1bii	5	95		
Q2a	2	72	26	
Q2b	19	7	74	
Q2c	83	17		
Q2d	40	60		
Q3a	8	92		
Q3b	40	60		
Q3c	40	60		
Q3d	9	91		
Q3ei	38	62		
Q3eii	13	10	14	64
Q4ai	37	63		
Q4aii	28	72		
Q4aiii	74	26		
Q4bi	36	64		
Q4bii	43	57		
Q5	22	34	29	15
Q6a	62	38		
Q6bi	75	25		
Q6bii	33	67		
Q6biii	16	5	25	55
Q6biv	33	22	45	
Q7ai	15	22	9	54
Q7aii	40	60		
Q7bi	53	47		
Q7bii	13	33	55	
Q7c	33	67		
Q7d	14	86		
Q8a	35	65		
Q8b	47	53		
Q8c	65	35		
Q8d	33	67		
Q8e	68	32		
Q9a	28	72		
Q9b	29	71		
Q9ci	87	13		
Q9cii	57	43		
Q9ciiiA	28	72		
Q9ciiiB	57	43		
Q10a	38	62		
Q10bi	31	69		
Q10bii	48	52		
Q10biii	57	43		
Q11ai	42	58		
Q11aii	81	19		
Q11bi	10	90		
Q11bii	9	91		
Q11c	36	6	19	39
Q12ai	12	88		
Q12aii	20	80		
Q12aiii	14	86		
Q12b	36	64		
Q12ci	36	64		
Q12cii	67	33		
Q13	24	38	24	14